Emergency AI co-scientist — project overview and plan

Prepared 20 October 2026 for discussion on Thursday 22 October. Week 4 of 10; the final week is 30 November – 4 December.

This document is the project overview and plan agreed in our last conversation. It is written to be read in one sitting and to be argued with. Three companion documents carry the detail: *ARCHITECTURE.md* for the system, *W4_DEBRIEF_AND_TARGET_SELECTION.md* for the October results and target analysis, and *output/TEN_WEEK_DELIVERY_PLAN.md* for the 82-task register with dependencies, owners and acceptance criteria.

Nothing below asserts binding affinity, functional inhibition, antiviral activity or therapeutic efficacy. No macrocyclic peptide and no non-canonical amino acid has yet been measured or modelled in this project. Where a number appears, §4 states what it was measured on and what it does not transfer to.

Summary

The project builds an **emergency AI co-scientist**: a system that helps experts decide what to pursue, what to measure next, and how to revise as evidence arrives, under the conditions that actually characterise an outbreak — incomplete pathogen knowledge, conflicting evidence, hard deadlines, finite compute, and very limited experimental capacity. Macrocyclic peptide discovery on SynX's mRNA display platform is the application and the experimental testbed.

Four weeks in, the position is this.

The research question has sharpened. It began as "can an agent make better scientific decisions under emergency constraints?" The October work suggests a more specific and more defensible claim: *the binding constraint on scientific-decision agents is not policy architecture but the absence of verifiable decision environments at the volume needed to train and evaluate one* — and public deep mutational scanning (DMS) landscapes are exactly such environments, already existing in quantity, and not currently used that way. Every variant's outcome is known, so a proposed measurement, ranking or stopping decision can be graded deterministically with no human and no wet lab. §1.3 and §2.2 make the argument.

Three of the project's headline claims failed this week, including one I made yesterday. The eightfold recovery over random selection is an artefact of pool size. The selected-versus-returned ratio — until this morning the one result I thought had replicated everywhere — is an accounting identity rather than a measurement: a feedback-blind policy reproduces the whole band, and the matched comparison of delayed against undelayed feedback is null. And my own claim from yesterday, that the gain tracks combinatorial

encoding rather than pathogen identity, was a mis-attribution with a circular control group. Each is now a permanent check in the code rather than a note in a document. §4 gives the arithmetic and §4.3 says what replaced them.

What survives is better than what it replaced. Acquisition gain is governed by whether the surrogate can learn *inside the candidate pool* — a computable property of pool geometry rather than of the biology, established by intervention in both directions rather than by correlation. It then replicated on two unrelated proteins against a prediction committed before the replays ran: predicted +0.25 and +0.24, observed +0.254 and +0.254, with every falsifier passed. Across eight landscapes on complete sweeps the groups do not overlap and the exact permutation test is at its floor, $p = 1/70$. The practical consequence is favourable: the closest on-disk analogue to a randomised display library gives the largest gain measured anywhere in the work, from a deliberately weak surrogate. §4.2 states the limits, which are real.

The system now holds biology. Until 19 October the software chose among three abstract strategies over numbered candidates and contained no representation of the problem domain. It now operates on a seven-axis target dossier with per-claim provenance, refuses evidence dated after the decision under consideration, retains contradictory evidence alongside the claim it contests rather than overwriting it, and states what a single selection round can and cannot establish. §3 and *ARCHITECTURE.md* describe it.

Both families you endorsed are adopted. Coronaviridae via SARS-CoV-2 Mpro and Orthomyxoviridae via the haemagglutinin stem or neuraminidase, with Nipah retained as a stretch robustness case. The deciding consideration was public measurement data, which lets method development proceed without a dependency on SynX. §5 sets out the rubric and §5.4 responds to your point on target identification. The prototype-versus-priority pathogen comparison you suggested is in *PROTOTYPE_AND_PRIORITY_PATHOGENS.md*.

Three external dependencies are overdue and none has an internal workaround. A named SynX experimental owner with a booked slot; the campaign export, without which the mandatory platform-improvement deliverable cannot begin; and confirmed trainable-model access with an approved spending ceiling. §8 states the consequence of each, and §10 asks for a decision on each.

On visibility. You are right that the output has been sitting locally. §9 proposes a concrete fix: repository access, a fixed weekly written cadence, and a standing rule that every claim in a document names the file and run that produced it.

1. The problem and the research question

1.1 What "emergency" means operationally

The word does a lot of work and is usually left vague, so it is worth pinning down. Five conditions distinguish outbreak-time research from ordinary research, and the project treats each as a thing to be represented in the system rather than as background colour:

- **Incomplete knowledge of the pathogen.** At the point decisions matter most, there may be a genome and little else — no structures, no assays, no reagents, no literature.
- **Evidence that evolves and conflicts.** Preprints contradict each other; early reports are revised; a claim that was reasonable on Tuesday is wrong by Friday. A system that overwrites its prior belief loses the information that the question was ever contested.
- **Hard deadlines.** The value of a decision decays. A correct answer in twelve months is worth less than a defensible answer in three weeks.
- **Finite compute and tool cost.** Structure prediction, molecular dynamics and large-model inference all cost money and time that is being spent instead of something else.
- **Very limited experimental capacity.** This is the binding one. A laboratory can run a small number of selection rounds. Choosing what to put into them is the decision that dominates the outcome.

The last two mean the central object is not a prediction but an **allocation**: given what is known now, what is the next action worth paying for? That framing is what makes the problem tractable as a measurable research question rather than an aspiration.

1.2 The question

> Can an AI co-scientist make better, faster, evidence-backed scientific decisions under emergency constraints, and revise appropriately as evidence and experimental results arrive?

Three levels of evaluation follow from it, and they are deliberately kept separate so that a failure at one does not contaminate the others:

1. **Scientific decisions (primary).** Target prioritisation, next-action choice, uncertainty handling, adaptation to evidence, and decision quality against time and cost.
2. **Molecular tools (supporting).** Reliability of structure, design and scoring tools; molecular ranking; chemical and structural transfer.
3. **End-to-end validation.** A historical COVID-19 reconstruction from January–March 2020 evidence, two prototype-family case studies, and the required prospective display round — with conclusions limited strictly to what was actually measured.

1.3 Where the contribution sits

A tool-using agent coupled to a structure-prediction stack is not by itself a novel contribution; Agent Rosetta already couples an LLM to Rosetta and evaluates non-canonical design. The project needs a sharper claim, and the October work points at one.

Building and evaluating a decision policy requires **environments in which decisions can be graded**. Expert review is the obvious grader, but it is slow, expensive, and has inter-rater-reliability problems that are themselves a project risk. The alternative is an environment where the ground truth is already known and the grading is exact.

Public DMS landscapes are that. A landscape such as `R1AB_SARS2_Flynn_2022` records a measured outcome for each of 5,725 Mpro variants. Hide the outcomes behind an oracle, expose only a tool API, let a system request measurements under a budget, and every decision it makes can be scored deterministically. Vary pool composition, budget, feedback delay, initial evidence, corrupted evidence and conflicting evidence, and one landscape yields an effectively unbounded supply of graded episodes. Ten such landscapes are already on disk (§5.2).

The claim is therefore: **auto-gradeable scientific decision environments can be manufactured at volume from existing public measurement data, and doing so is what makes a trainable, evaluable decision policy possible.** The emergency co-scientist is the system that demonstrates it; the SynX display round is the prospective check that the decisions survive contact with a real laboratory.

This also rescues the October pilot work. The GB1 study was built as a biology pilot, which is why it looks thin as biology. As infrastructure it is the first working instance of exactly the pattern above.

2. Why the project has this shape

This section is the "reasoning around why you made certain choices" you asked for. Each subsection states a decision, the alternatives, and what would make me reverse it.

2.1 Why decision quality rather than molecular prediction

The choice. The primary measured object is the quality of the *decision procedure*, not the accuracy of a molecular predictor.

The reasoning. Acquisition policy produces gains over random selection *while the underlying predictor is only weakly informative* — Spearman ≈ 0.31 at a 128-label budget, with plain ridge regression matching a 64-tree forest. If decision gains had required a good predictor the emergency premise would collapse, because at emergency timescales there are no labels with which to fit one.

How much of that survived this week. The weak-predictor half is intact and better supported: the largest gain in the project comes from exactly that Spearman-0.31 surrogate. The half that does not survive is the stronger reading — that gains come from the policy *rather than* the model, and are therefore substrate-independent. They are a surrogate effect; where no weak model can be fitted inside the pool, no policy beats random. §4.3 gives the evidence and the corrected claim.

What this costs and what it buys. It costs the cleanest version of the argument. It buys a precondition that can be *computed in advance* — pool coverage — which is more useful to a decision-maker than a claim of substrate independence would have been, because it says when to expect the method to work before any budget is committed.

What would reverse the remaining claim. If the gain fails to appear on two independent high-coverage landscapes staged from unrelated proteins — the next experiment, §10 — then the mechanism rests on one independent protein plus a restaging of the pilot, and the contribution narrows to the evaluation-design work and the environment generator. I would rather reach that in week 6 than at review.

2.2 Why public measurement landscapes as the substrate

The choice. Method development runs on public DMS data rather than on SynX campaign data.

The reasoning. Three things follow from it. First, **it removes a blocking dependency**: the SynX campaign export has not arrived, and a plan whose method development cannot start without it is a plan with a single point of failure. Second, **the outcomes are already known for every candidate**, which is what makes automatic grading possible at all. Third, **it is releasable**: code, benchmark definitions and evaluation data can be published without negotiating permissions, which matters for the artifact deliverable and for the paper.

The cost, honestly. Public landscapes are single-substitution scans of canonical residues. They contain no macrocycle, no non-canonical chemistry and no enrichment readout — the two features that define the SynX platform. No result from them can speak to either. The landscapes are a decision-environment substrate, not a model of the platform, and §4.3 keeps that boundary explicit.

What would change it. The arrival of the campaign export, which would let the same episode machinery run over real macrocycle measurements. The request is live and overdue.

2.3 Why a target dossier as the decision state

The choice. The system's state is a seven-axis target dossier with per-claim provenance, not a feature vector.

The reasoning. Before 19 October the state was `observed_score_summary`, `candidate_summary[{candidate: int, prediction, uncertainty, diversity}]`, `remaining_label_budget`, `pending_labels`, `feedback_delay_rounds`, `decision_round`, with an action space of `{greedy, explore, diverse}`. That is a bandit-strategy selector. It reasons competently about budget and ignorance and contains no biology whatsoever, and it is several abstraction layers away from what the project is supposed to deliver: pathogen intake from genomic sequence → evidence retrieval → structure prediction → target and epitope ranking → library design.

Three properties of the dossier are deliberate and are the ones I would defend hardest:

- **Per-claim provenance.** Every assertion names its source. An assertion with no source is marked as requiring verification rather than being silently trusted — the system currently flags its own claim about prior macrocycle campaigns against Mpro as uncited.
- **Temporal refusal.** Evidence dated after the decision under consideration is rejected. This is what makes the historical COVID-19 replay meaningful rather than a demonstration of hindsight.
- **Conflict retention.** Contradictory evidence sits alongside the claim it contests, with both timestamps, rather than overwriting it. An emergency decision-maker needs to know that a question is contested; a system that overwrites loses exactly that.

The third has an under-appreciated cost: the dossier grows monotonically within an episode, and what gets evicted from it becomes a decision the policy itself has to make. That is an open design question, named in §10.

2.4 Why macrocyclic peptides and mRNA display

This was given by the project context rather than chosen, but it is a good fit and the reasons are worth stating because they shape target selection.

RaPID-style mRNA display generates libraries of order 10^{12}–10^{13} by flexizyme-charged initiator incorporation with spontaneous thioether macrocyclisation. The chemistry is largely target-independent, which means **the reagent is the binding constraint, not the library** — a target is tractable if SynX can

obtain soluble, correctly folded, immobilisable protein that survives several selection rounds, and not otherwise. That single fact dominates the selection rubric in §5.1.

It also creates a tension the project should own rather than finesse: the best-behaved viral reagents are soluble intracellular enzymes, which carry a membrane-permeability liability; extracellular glycoproteins need no permeability argument but are harder reagents and sit under strong positive immune selection, which undermines conservation-based family claims. Choosing a protease and owning the permeability problem is the more defensible position, because permeability is predictable in silico with tools already in the bibliography whereas conservation of an immune-evasion surface cannot be engineered around.

2.5 Why these two families

Coronaviridae via Mpro and Orthomyxoviridae via the haemagglutinin stem or neuraminidase, per your endorsement. Beyond that endorsement the pair was chosen because:

- **Both have public labelled data**, which unblocks the molecular transfer tests immediately. The H1→H3 grouped holdout — the exact distribution-shift test the brief specifies — is runnable on data already on disk.
- **Together they span the §2.4 tension deliberately.** One soluble intracellular enzyme with a permeability problem; one conserved extracellular epitope without one. That makes the pair a designed contrast rather than two arbitrary picks, which is a better case study structure.
- **Nipah is retained as a stretch robustness family.** Better narrative — WHO priority pathogen, genuinely extracellular, a real conservation ladder for transfer testing — and no public DMS, so it cannot carry method development.

2.6 What was considered and set aside

Recording these matters as much as recording what was chosen.

- **The structural arm is retired as a scientific component.** It measured how fast mean per-residue solvent-accessible surface area converges to a held-out MD replicate under different allocation policies. The effect was real but small, and the deeper problem is that agreement with a sibling simulation is not an epitope-accessibility measure in any sense that feeds target ranking. Nothing in the theory of change consumes it. Re-pointing it at viral glycoproteins would not fix that. The cost-accounting and evidence-isolation code is kept.
- **Spike RBD was rejected as the family narrative target** despite having the cleanest binding landscape available. It is an immune-evasion surface under intense positive selection and is therefore the wrong choice for any conservation or family-transfer claim. It stays as a method-development substrate only.
- **The SARS-CoV-2 frameshift stimulatory element** — a structured-RNA target with mRNA display precedent in the project's own bibliography — is the most novel option available and sidesteps the conservation problem entirely, because RNA secondary structure is under stronger constraint than a

glycoprotein surface epitope. Against it: non-standard reagent and assay, uncertain functional readout, no public landscape. Worth ten minutes on Thursday even if declined.

3. System architecture

ARCHITECTURE.md is the full description with module-level detail. This is the shape and the rationale.

3.1 Layers

The system is a Python package, `hpc_pilots`, with a single CLI entry point and a strict separation between four concerns:

LAYER	RESPONSIBILITY	WHY IT IS SEPARATE
Domain	Target dossiers, pathogen target packets, the non-canonical building-block registry	The biology must be representable independently of any policy that reasons over it. Writing the dossier schema *before* the policy that consumes it was deliberate; the reverse order produces a schema shaped by whatever the first policy happened to need.
Environment	Decision episodes built from measurement landscapes: hidden outcomes, timed evidence release, budgets, deadlines, delayed feedback	This is the auto-gradeable oracle of §1.3. It owns the hidden truth and exposes only a request interface.
Policy	The rules being compared: random, greedy, UCB, diversity-seeking, and an LLM-mediated policy over the six-action space	Policies are interchangeable behind one interface so that comparisons are matched by construction rather than by care.
Evaluation	Scoring, calibration studies, summarisation, provenance and resource accounting	Kept out of the policy path entirely, so a policy cannot read a metric it has not earned.

3.2 The isolation boundary, which is the part that matters

The single most important property of the system is that **a policy cannot reach an answer it has not earned**, and it is enforced structurally rather than by convention:

- The full outcome vector never leaves the evaluator. The policy interface accepts only revealed values, enforced at the function boundary in `pick_batch` and covered by a test.
- Top-fraction membership is never supplied to a policy.
- Candidate identities in LLM prompts are opaque integer pool indices, so a model cannot recognise a variant and recall its phenotype from pretraining.
- For the structural data, one entire MD replicate is held out and used only inside the evaluator.

This is the same boundary that the containerised oracle of §7 generalises. Building it first was the right call: an evaluation harness whose isolation is a matter of discipline rather than structure will leak, and a leak discovered in week 9 invalidates everything after it.

3.3 Reproducibility, and why it was worth the time

Roughly a fifth of the engineering effort went into provenance machinery. The justification is that a result I cannot regenerate is a result I cannot defend at review:

- SHA-256 hashes for every downloaded input and every frozen split, in `data/processed/*/manifest.json`.
- Every result file embeds `dataset_manifest_sha256` and `config_sha256`, so any result ties back to the exact configuration and data that produced it.
- Atomic JSON writes with `allow_nan=False`, so a NaN cannot silently enter a results file and a partial write cannot be mistaken for a result.
- Smoke-test outputs carry a `smoke` flag and are excluded from scientific summaries by construction, not by remembering to filter them.
- An inference or analysis failure fails its task explicitly. **A failed LLM call never degrades into a baseline policy result** — this one has caught real bugs.
- Seeded tie-breaking before a stable argsort. This sounds like trivia and is not: the GB1 landscape has large exactly-tied plateaus at low budget, and unseeded tie-breaking silently injects a diversity bonus into the greedy policy.

The calibration studies regenerate from committed scripts. One column does not reproduce bit-exactly and this is recorded rather than hidden: the original ridge penalty was not recoverable, so the committed script uses `RidgeCV` over a log-spaced grid, reproducing 31 of 160 rows exactly with mean absolute difference 0.0022 on Spearman. No quoted figure changes.

3.4 Model access and compute

The LLM policy speaks to any OpenAI-compatible backend selected by `PILOT_LLM_PROVIDER` — local vLLM, OpenRouter, Anthropic or OpenAI — with per-provider request shaping. Actions are constrained by a strict JSON schema; an action outside the allowed set, or a reply containing no JSON object, raises rather than falling back. Hosted runs get a provider-specific policy label so they can never pool with or overwrite local-model runs.

Compute runs on the MATS cluster under SLURM, with dependency-gated job chains and a submission ledger so that scheduler state is recorded from scheduler evidence rather than inferred from the fact that a job was staged.

4. What has actually been measured

Four weeks of evidence, separated into what has been withdrawn, what survives, and what has not been attempted at all. This section supersedes §2 of the *Week 4 debrief*, which is left unrevised as the record of what was believed on 19 October and carries a correction notice at its head. The intervention tests behind §4.2 are set out in *ADJUDICATION_COMBINATORIAL_GAIN.md*, and the replication prediction in *PREREGISTRATION_COVERAGE.md*.

4.1 Three claims withdrawn

I am leading with these rather than burying them. Each was a number I believed, two of them appear in the Week 4 debrief you may already have read, and one of them I generated yesterday. All three are now permanent checks in the code, so they cannot quietly return.

The eightfold gain over random (withdrawn 19 October). Random top-fraction recall is simply the labelled fraction of the pool, so an improvement-over-random is bounded by pool geometry: 32× of headroom at 128 labels from 4,096 candidates, but only 4× at 128 from 512. The figure measured the pool, not the policy. A related error was found yesterday in the correction itself: the attainable multiple is `min(1/top_fraction, 1/labelled_fraction)`, not `1/labelled_fraction`. The two agree at the final checkpoint of every configured landscape, so no reported figure changes, but the old formula overstated headroom precisely at the early checkpoints whose raw multiples look most impressive.

The selected-versus-returned ratio (withdrawn 20 October). This was the project's best result and the one I was most confident in. It is an identity. `returned` is always a subset of `selected` and recall divides both by the same pool-wide hit count, so the ratio is monotone in set inclusion — across 1,018 recorded delayed checkpoints it never once falls below 1. What it reports is the share of the budget still in flight: with 128 labels in batches of 8 at a three-round delay, 24 are pending at the final checkpoint and 128/104 = 1.23 falls out of the bookkeeping before any policy acts.

The decisive check is the random policy. It reads no returned label, so its selections are identical at every delay — confirmed in 99 of 99 pools — and yet it reproduces the entire band:

	MEAN RATIO	SD	N
Random (feedback-blind)	1.297	0.498	89
Greedy / UCB / diverse (informed)	1.312	0.356	279

A quantity a feedback-blind policy can produce is not a measurement of how delay degrades a decision. And the matched comparison that *would* measure it — same budget, same metric, delay 0 against delay 3 — is null: mean paired difference −0.006 across 392 pool comparisons, Wilcoxon $p = 0.31$ on the 243 non-zero pairs. **The 1.1–1.4× figure must not be presented as a finding.**

That the gain tracks combinatorial encoding (withdrawn 20 October). Mine, from yesterday's runs. Two failures. The control group was circular: `spike_rbd_ace2` *is* SARS-CoV-2 spike RBD, so across the six staged landscapes "viral protein" and "single-substitution scan" are the same two-versus-four partition, and any statistic separating one separates the other identically. And the explanatory variable was wrong — combined substitutions are collinear with, but not the cause of, the quantity that actually moves the gain. Intervention breaks it in both directions (§4.2). It is the same failure mode as the eightfold figure: a property of pool geometry read as a property of the biology.

4.2 What survives — a mechanism, with an intervention test

Acquisition gain is governed by **how densely the candidate pool covers the variable positions the surrogate must learn from** — that is, by whether a weak model can be fitted *inside* the pool at all. This is a computable property of pool geometry, available before any assay is run.

Correlation would not establish that, so it was tested by intervention in both directions, holding everything else fixed:

INTERVENTION	GREEDY GAIN	SURROGATE SPEARMAN
Influenza H3 scan, pools restricted to 28 variable positions	+0.104 → **+0.250** (8/8 pools)	0.10 → 0.63
…to 55 / 150 / 566 positions	+0.159 / +0.019 / 0.000	0.42 / 0.18 / 0.10
GB1 doubles re-encoded so pool members share no feature levels	+0.242 → **−0.019**	0.66 → 0.03

The first takes a pathogen protein on a replication-fitness endpoint — the very case that appeared to fail — and recovers the full gain by narrowing the pool. The second keeps the same variants, the same hidden scores and the same combinatorial structure, destroys only the feature sharing, and removes the gain entirely. Combined substitutions are neither necessary nor sufficient.

It is a *surrogate* effect: the surrogate-free diverse policy shows no systematic gain on any landscape (−0.042 to +0.058 across all eight), and it requires at least 64 of 512 labels — absent at 8, 16 and 32.

It then held on two unrelated proteins, against a prediction written down first. The mechanism's weakest point was independence: it rested on one independent protein plus a restaging of the pilot protein. Two further landscapes were staged — GRB2 SH3 domain and KRAS against the DARPin K55 binder, neither related to anything already in the set, the second carrying a binding rather than a fitness endpoint. The predictor (in-pool surrogate Spearman at 128 labels), the thresholds and the falsifiers were recorded in *PREREGISTRATION_COVERAGE.md* and committed **before** the replays ran. Predicted ≥ +0.15 improving in ≥ 7 of 10 pools, with point estimates of +0.25 and +0.24; observed +0.254 and +0.254. Every criterion passed, including diverse staying within ±0.05 of zero on both.

Running every staged pool — no cell now rests on a selected half — the comparison is:

LANDSCAPE	IN-POOL SPEARMAN	POOLS	GREEDY GAIN	POOLS IMPROVED
GRB2 SH3	0.46	20	**+0.265**	19/20
YAP1 WW	0.35	18	**+0.263**	17/18
GB1-Olson (matched)	0.70	20	**+0.244**	17/20
KRAS–DARPin	0.68	20	**+0.194**	15/20
Mpro	0.09	10	+0.065	7/10
Influenza HA H3	0.08	20	+0.060	13/20
Spike RBD	0.11	6	+0.030	3/6
Influenza HA H1	0.12	19	−0.004	10/19

The groups do not overlap — lowest high-coverage +0.194 against highest low-coverage +0.065 — and the exact one-sided permutation test over landscape labels now gives **p = 1/70 = 0.014**, against a floor of 0.067 before these two were added. Finishing the partial sweeps was not cosmetic: H3 fell from +0.104 on ten pools to +0.060 on twenty, and KRAS from +0.254 on ten to +0.194 on twenty.

The relevant analogue for this project is better than either. The four-site GB1 library (Wu 2016) is a randomised region over an alphabet, every member carrying a value at all four randomised positions, members mutually distant (mean pairwise Hamming 0.95). At the same 512/128/5% geometry it gives **the largest gain measured anywhere in this work — greedy +0.373 and UCB +0.373, 10 of 10 pools — from a surrogate with Spearman 0.31.** What matters is coverage of the randomised positions, not proximity to a wild-type reference, and a randomised macrocycle region of a few positions sits at the favourable extreme of that variable.

Six things that must travel with this.

- It is an argument about pool structure, not a measurement of macrocycles. No landscape here contains a macrocycle, a non-canonical residue or a cyclisation.
- Coverage is necessary but not sufficient. Mpro reaches Spearman 0.68 with greedy gain −0.02; GB1 singles reach 0.61 with no gain. The surrogate must also resolve the top tail. Two conditions of unknown relative weight, not a rule.
- The gain decays steeply with pool size at fixed budget: normalised gain 0.58 → 0.22 → 0.09 → 0.035 at 512 → 2,048 → 8,192 → 32,768 candidates. It therefore describes focused pools of hundreds to a few thousand — deconvolution and focused maturation — and says **nothing** about a selection round over a 10^{11}-member library. Over that same sweep the *multiple* over random rises from 2.9× to 12.9× while the policy loses almost all its real advantage, which is exactly how the eightfold figure misled. No multiple over random will be quoted again.

- Eight landscapes is still eight. The permutation test is at its floor (rank 1 of 70), so 0.014 is the smallest p this design can produce, not a measurement of how much smaller it might be. The ± spreads quoted per landscape are variation across pools of one protein, not evidence about the landscape class.
- **This does not separate coverage from combinatorial structure.** Both new landscapes are multi-substitution and high-coverage, so the withdrawn hypothesis predicts the same outcome. It tests independence only. Separating the two needs the within-protein comparison — PDZ doubles against the already-staged PDZ singles with coverage matched — which is not yet run.
- Coverage is also not the whole story in the other direction: KRAS has the second-highest in-pool Spearman (0.68) and the smallest gain of the four (+0.194), and gain is not monotone in surrogate skill anywhere in the table.

4.3 What this does to the project's central argument

The project has argued that decision gains do not require a good predictor — which matters, because at emergency timescales there are no labels with which to fit one. That argument **partly survives, in a more useful form.**

What survives, now better supported: gains do not require a *good* predictor. The largest gain in the project comes from a surrogate at Spearman 0.31, and gain is not even monotone in surrogate skill — blocked H3 reaches 0.63 with a smaller gain. A modestly informative model is enough.

What does not survive: the stronger reading that gains come from the policy *rather than* the model, and are therefore substrate-independent. They are a surrogate effect. Every landscape where the surrogate has essentially no in-pool skill shows essentially no gain, and the surrogate-free policy never gains anywhere. The original argument was measured on the four-site GB1 library — the highest-coverage substrate in the project — so "substrate-independent" was never tested where coverage is poor. It has now been tested four times and it fails there.

The honest version, and the one I would defend on Thursday: *policy gains require only a weakly informative surrogate, but they do require one — and whether one can be fitted at all is a property of the candidate pool's geometry that can be computed in advance.* That is weaker than the original and considerably more useful, because it is checkable before any budget is spent.

4.4 Why I am presenting it this way

Three self-caught errors in two days is not a comfortable thing to put in front of you, and the temptation to lead with the surviving mechanism instead was real. But the project's proposed contribution is that auto-gradeable decision environments let you find your own artefacts fast, and this week is the first evidence that the claim is true of the method rather than just asserted about it. Each correction is now a control in the code — a delay-effect table carrying the identity and the null band beside the effect, a corrected headroom ceiling, a permuted-outcome falsification run, and a zero-retrieval control for the dossier arm — so the specific mistakes cannot recur silently. That is the part worth carrying into the paper.

4.5 What has not been measured at all

- **No macrocycle, and no non-canonical amino acid.** The building-block registry carries literature block *classes* only; every chemical identifier, codon assignment, Rosetta parameter and rotamer library is null pending the campaign export.
- **The LLM policy has no non-smoke result anywhere.** Only the four heuristic baselines have scientific runs on disk. The emergency episodes have been run with deterministic policies only, and those are stable under irrelevant evidence by construction, so the stability metric has not yet been tested against anything that could fail it. This is week 5 work and it is the most important near-term gap.
- **The physics-based non-canonical track is untouched** — Flex ddG, rotamer library generation, the RosettaCommons non-canonical interface protocol. Nothing in the repository approaches it.
- **No prospective experimental measurement**, which is what the display round is for.

5. Targets

5.1 The selection rubric

Seven axes, which are simultaneously the selection criteria and the schema of the decision state in §2.3:

AXIS	WHAT IT ASKS	WHY IT GATES
Reagent tractability	Can SynX obtain soluble, correctly folded, immobilisable protein surviving multiple selection rounds?	Hard gate. An untractable reagent ends the campaign regardless of biology.
Essentiality	Does loss of function block replication?	A binder to a dispensable protein cannot become a therapeutic.
Conservation	Is the epitope conserved across the family, and under what selective pressure?	Gates every family-wide and prototype-pathogen claim.
Accessibility	Intracellular or extracellular; is the epitope exposed in the relevant conformation?	Determines whether a permeability argument is required.
Assay availability	Is there a functional orthogonal assay beyond enrichment?	Without one, a single round substantiates enrichment and nothing more.
Public label availability	Is there a DMS or binding landscape for method development?	Determines whether transfer tests run now or block on SynX.
Precedent	Prior macrocycle or peptide campaigns against this target?	Supplies positive controls and de-risks failure interpretation.

5.2 The recommendation, and the data behind it

Coronaviridae via Mpro; Orthomyxoviridae via HA stem or NA; Nipah as stretch.

Mpro (3CLpro, nsp5) scores strongly on reagent tractability — routine soluble *E. coli* expression, well behaved, inexpensive — on essentiality, on active-site conservation across sarbecoviruses, and on assay availability via the standard FRET cleavage assay. It is weak on accessibility, being cytosolic, and so requires a permeability argument. That liability may be an asset for this project: it forces a genuinely multi-objective decision, affinity against predicted permeability, which is a harder and more interesting problem for a co-scientist than single-score ranking.

Ten pathogen landscapes are already on disk:

DMS ID	TARGET	MUTANTS
`R1AB_SARS2_Flynn_2022`	SARS-CoV-2 Mpro	5,725
`SPIKE_SARS2_Starr_2020_binding` / `_expression`	SARS-CoV-2 Spike RBD	3,802 / 3,798
`A0A2Z5U3Z0_9INFA_Doud_2016`	Influenza HA (H1N1)	10,715
`C6KNH7_9INFA_Lee_2018`	Influenza HA (H3N2)	10,754
`NRAM_I33A0_Jiang_2016`	Influenza neuraminidase	298
`A4D664_9INFA_Soh_2019`	Influenza PB2	14,421
`RDRP_I33A0_Li_2023`	Influenza PB1	12,003
`POLG_DEN26_Suphatrakul_2023`	Dengue NS5	16,897
`A0A140D2T1_ZIKV_Sourisseau_2019`	Zika envelope	9,576

For the single display round the proposal is Mpro: the least expensive well-behaved reagent, the strongest conservation case, an orthogonal functional assay if scheduling permits, and interpretable failure. One round substantiates enrichment and must be reported as enrichment.

5.3 Prototype and priority pathogens — and an uncomfortable finding

The comparison you suggested is in *PROTOTYPE_AND_PRIORITY_PATHOGENS.md*, covering WHO, UKHSA, NIAID, CEPI, HERA, the US operational and regulatory lists, other national frameworks and the academic literature. Three things from it belong here rather than in a companion document.

The two kinds of list answer different questions and are not two tiers of one ranking. A *priority* pathogen is selected by risk: how bad, how likely. A *prototype* pathogen is selected by representativeness and tractability: if we build tools against this one, do they transfer to its relatives, and can we work on it today? The mismatch runs both ways and is explicit in the sources. WHO names vaccinia a Poxviridae prototype partly because it offers a BSL-2 route, and *Vesiculovirus* a Rhabdoviridae prototype because it is a useful vaccine vector — tool-value criteria applied to a family WHO rates low risk. Conversely variola is a priority pathogen and explicitly not a prototype. A prototype list is not a ranking of danger, and citing it as one inverts its meaning.

NIAID excluded both of the families we chose, by name, for a reason that applies to us. I verified this against the primary source rather than taking it from the survey. The NIAID workshop that operationalised the prototype approach covered ten families — Arenaviridae, Hantaviridae, Nairoviridae, Peribunyaviridae, Phenuiviridae, Filoviridae, Flaviviridae, Paramyxoviridae, Picornaviridae and Togaviridae — and states:

> "Although Orthomyxoviridae and Coronaviridae have a very high pandemic risk, they were not included in this prototype pathogen exercise since significant resources are already allocated to characterize these families, and there are licensed or advanced clinical stage development MCMs available for viruses within them."
>
> — Deschamps, DeRocco, Bok & Patterson, *J Infect Dis* 2023;228(Suppl 6):S355–S358

NIAID's criteria are *inverse* to resourcing: the better-studied a family, the lower it ranks as a prototype. Our selection criterion — public measurement data exist — is close to the exact quantity that caused the exclusion. A reviewer aligned with NIAID will not ask whether the choice is risky enough; they will ask why not Nairoviridae or Picornaviridae, where the field is demonstrably unprepared and where ReVAMPP actually spends.

That is not fatal, and the answer I would give is that NIAID's exclusion concerns *product* resourcing for countermeasure development, not benchmark availability for *method* validation — a family with dense public quantitative data is the right place to validate a method whose intended destination is the data-poor families. But that answer is only credible if the method is then shown to transfer out of distribution, and if the data-poor destination is stated as the goal rather than left implicit. It is worth ten minutes on Thursday.

Two corrections to how the project states its targets. WHO prioritises Coronaviridae at **subgenus** level, so the Mpro conservation argument should be made across *Sarbecovirus* and *Merbecovirus*, not against SARS-CoV-2 alone — which is a stronger claim and the one WHO's own molecular rationale invites. And "influenza haemagglutinin stem" is not a WHO unit: WHO prioritises influenza by HA subtype. The project needs to say which subtype, and whether the conservation claim is within an HA group or across groups. Both are free improvements to the framing.

5.4 On the target-identification analogy

You wrote that the analogue of target identification for a novel pathogen would be identifying targets across novel disease biology, and that this is not useful because pathogen-relevant biology differs from other disease biology.

Taken, and the question is withdrawn. The practical consequence is that the project does **not** pursue a general target-identification method transferable across disease areas. Target identification here is pathogen-specific and proceeds from the seven axes in §5.1, which are mostly virology-specific: polyprotein processing essentiality, conservation under immune pressure, the reagent behaviour of a viral protease versus a class-I

fusion glycoprotein. The transfer claim the project does make is the narrower and better-evidenced one — *within* a viral family, from a prototype to its relatives — which is the same claim the prototype-pathogen literature makes, and §5.3 examines how well it is actually evidenced.

6. Evaluation design

Four system-level studies, each with a decision attached, so that a negative result changes the plan rather than being absorbed:

STUDY	QUESTION	DECISION IT FORCES
S0	Do deadlines, incomplete evidence and limited capacity materially change which actions are useful?	If constraints do not affect the task, the emergency-specific contribution is not real and the benchmark is revised before any claim is made.
S1	Does structured evidence tracking beat simpler workflows on support, uncertainty and dossier quality?	If the agent offers no gain over a scripted workflow, identify the missing capability before adding orchestration.
S2	Does the system revise when evidence warrants and hold steady when it does not?	A changed ranking is not automatically a better ranking; this separates evidence-following from instability.
S3	Does an adaptive policy use a constrained budget better than a fixed action order?	Pursue the policy contribution only if gains survive held-out episodes and a simple cost-aware baseline.
S4	Do the recommendations survive a real laboratory?	The prospective check; predictions frozen before results arrive.

Four commitments govern all of them. **Baselines are matched** — a fixed rubric, a scripted workflow, retrieval with a single LLM response, and the adaptive agent, on identical document access, tool capabilities and molecular models. **Splits are grouped, not random** — by parent peptide, backbone and chemical similarity group, because a strong random-split result with weak grouped transfer supports an interpolation claim and nothing more. **Prospective predictions are frozen** before measurements arrive, with all submitted candidates, selection rules, failures and controls recorded. **Hindsight leakage is audited** — date-limited retrieval does not remove knowledge already in model weights, and the historical COVID-19 claims will state that limit explicitly.

7. The plan

7.1 Required outputs, and where each is satisfied

No stretch goal has been dropped. If a collaborator-dependent obligation fails it is recorded as incomplete rather than replaced with a simulation.

REQUIRED OUTPUT	WHERE IT LANDS	STATUS
Modular emergency co-scientist	Decision state and action space delivered 19 Oct; integrated replay W5; demo frozen W6	Partially delivered
Evidence-backed target dossiers	Seven-axis packets for three families exist; both chosen families finalised W5	Partially delivered
Non-canonical macrocyclic designs and focused library	Block registry exists as structure; chemistry blocked on campaign export	**Blocked**
Historical COVID-19 evaluation	W5, with hindsight audit	Planned
Two prototype-family case studies	W5–W6, scored outputs	Planned
SynX platform improvement	Blocked on campaign export; re-scope decision at Thursday's gate	**Blocked**
One mRNA display selection round, analysed	W7 execution; readout due 18 Nov; analysis by 20 Nov	**Blocked on slot**
Paper, reproducible artifact, presentation	W10	Planned

7.2 Remaining shape of the ten weeks

WEEKS	FOCUS	GATE
W4 · Oct 19–23	Development episodes; evidence-state ledger; family sign-off; round question	Feasibility and novelty gate, 23 Oct, with the export blocker named
W5 · Oct 26–30	LLM policy over the six-action space; both-family dossiers; historical replay; expert rubric	Platform-improvement choice, 27 Oct
W6 · Nov 2–6	Held-out system evaluation; **prospective predictions frozen 4 Nov**; London readiness	Readiness gate, 6 Nov
W7 · Nov 9–13	London: facility and design review, the actual selection round, expert feedback	Round executed
W8 · Nov 16–20	Training harness; throughput and spend gate; **lab readout due 18 Nov**	Throughput gate, 19 Nov
W9 · Nov 23–27	Forecasts saved before held-out reveal; analysis; independent review	**Results freeze, 27 Nov**
W10 · Nov 30–Dec 4	Clean reproduction, manuscript, artifact, rehearsals, final handoff	Handoff, 4 Dec

The register holds 82 tasks across eight streams and 273 estimated hands-on hours, of which 14 are delivered. A task reads `Planned` unless a status line names its evidence; only four currently do. The generator checks for missing and cyclic dependencies and for window ordering, and fails rather than emitting an inconsistent chart.

7.3 Capacity, stated plainly

Week 4 does not fit. Fourteen hours went into the domain layer on 19 October, which was not in the baseline. The remaining 38.5 hours still include the molecular audit path, which cannot start until the campaign export arrives. If the export has not arrived by 21 October the molecular path slides to week 5 and the mandatory platform improvement loses its slack. That is a blocker with an owner, not something to absorb silently.

Weeks 5 and 6 run at 38–40 hours against a 40-hour week — effectively no slack. Weeks 8 to 10 carry 6–12 hours of buffer each, deliberately reserved for API retries, late experimental QC and reproduction fixes. Week 7 is on-site review and observation; it is not additional coding capacity. Week 9 includes US Thanksgiving, so the results reviewer should be booked before London or the review pulled to 25 November.

8. Risks

RISK	CURRENT STATE	CONSEQUENCE	MITIGATION
No booked display round	Overdue since 16 Oct. No named SynX owner, no slot.	The one mandatory experimental deliverable does not happen. Backward planning from a usable 18 Nov readout is already tight.	Needs a name and a date this week. No internal workaround exists.
Campaign export not received	Overdue.	The platform-improvement deliverable cannot begin; the molecular audit path is entirely blocked. Most exposed dependency in the project.	Re-scope decision at Thursday's gate: either the export arrives, or the platform improvement is re-specified against something available.
Trainable model access unconfirmed	No confirmed account, model list or spend ceiling.	Cannot commit to model identities or an experiment matrix.	Verify live availability; the local pilot model is not automatically a supported training model.
The mechanism rests on few independent landscapes	§4.2. Now four against four with no overlap and a pre-registered replication on two unrelated proteins, but the permutation test sits at its floor ($p = 1/70$).	The claim is supported rather than established; a reviewer will ask for the within-protein control.	*Largely addressed on 20 October.* What remains is the 2×2 that separates coverage from combinatorial structure within one protein — PDZ doubles against the staged PDZ singles, coverage matched. ~60 replays, about two minutes.
Further headline claims may be artefacts	Three fell in two days.	Credibility, and wasted weeks building on a number that is not real.	The controls are now in the code rather than in a document: a delay-effect table carrying the identity and the null band, a corrected headroom ceiling, a permuted-outcome falsification run, and a zero-retrieval control. Every new claim gets an intervention test before it is written down.
Expert rubric reliability	Not yet measured.	If inter-rater reliability is poor, the primary evaluation is unreliable.	This is precisely why the auto-gradeable environments matter: they give a domain where ground truth is exact rather than assessed.
Scope	Eight workstreams, ten weeks, one person plus collaborators.	Thin delivery across everything rather than one solid result.	One paper narrative rather than two; mandatory outputs protected; a negative result retained and the claim narrowed, rather than the deliverable dropped.

9. Working arrangements — addressing the visibility problem

You are right that the output has been sitting locally, and that is a problem worth fixing properly rather than by sending more attachments. Proposed:

1. **Repository access.** The work is a git repository with full history — code, configurations, frozen splits, result manifests and these documents — and it already has a private GitHub remote, so this is one invitation away. Tell me which account to add, and whether anyone at SynX should have it too. I have not added anyone unilaterally because the repository will eventually carry campaign-derived material, and I would rather agree who sees it before that arrives than afterwards.
2. **A fixed weekly written update**, delivered the day before our call: what was decided, what was measured, what changed, what is blocked, and what I need. Short, and in the same honest register as §4.
3. **Claim-to-evidence traceability as a standing rule.** Every number in a document names the file and run that produced it, and anything asserted from memory is marked as requiring verification until a citation exists. There is one such flag outstanding right now — the claim about prior macrocycle campaigns against Mpro.
4. **Gate decisions in writing**, with the options, the recommendation and the reasoning, so a decision you make is recoverable later.

10. For Thursday

Ordered by how much they block.

Decisions I need from you

1. **The three overdue dependencies.** A named SynX experimental owner and a booked slot; the campaign export or an agreed substitute; confirmed trainable-model access with a spend ceiling. Each is past its date, each is outside my control, and the second and third both change the plan depending on the answer.
2. **The display-round question.** Mpro confirmed, or an alternative? This gates library design and everything downstream of it.
3. **Scope.** Eight streams in six remaining weeks is aggressive. If something should be cut rather than delivered thinly, Thursday is the time to say so.

Things I would value your judgement on

1. **Is the sharpened contribution the right one?** "Verifiable decision environments manufactured from public measurement data" versus "an agent that helps scientists decide." The first is more defensible and less obviously useful to a clinician; I think it is right but I would like to be argued with.
2. **Have I scoped §4.2 correctly?** My reading is that the surviving result describes focused pools of hundreds to a few thousand candidates — deconvolution and focused maturation — and says nothing about a selection round over a 10^{11}-member library. That is a deliberately narrow reading of my own result and I would rather you tell me it is too narrow than find out at review that it was too broad.
3. **NIAID excluded both our families from its prototype list, by name** (§5.3), because they are already well resourced — which is close to the exact reason we chose them. I have an answer, which is that their exclusion concerns product resourcing and ours concerns benchmark availability for method validation. I would like to know whether you find that convincing or whether it sounds like special pleading, because it is the one place where our target choice is in direct tension with a published framework rather than merely unsupported by one.
4. **The dossier eviction problem.** Evidence accumulates monotonically within an episode and conflicts are retained rather than overwritten, so context eventually has to be dropped — and what to drop is itself a decision the policy should be making. I do not have a good answer and it is a genuinely interesting design question.
5. **The frameshift element**, §2.6. Ten minutes, probably declined, worth asking.

What I will bring

The running system with a walkthrough of a single episode end to end; the replays behind §4, including the two interventions; and the arithmetic on all three withdrawn claims. If you want one thing to look at before Thursday, make it §4 — it is the most informative two pages about how this project actually works.